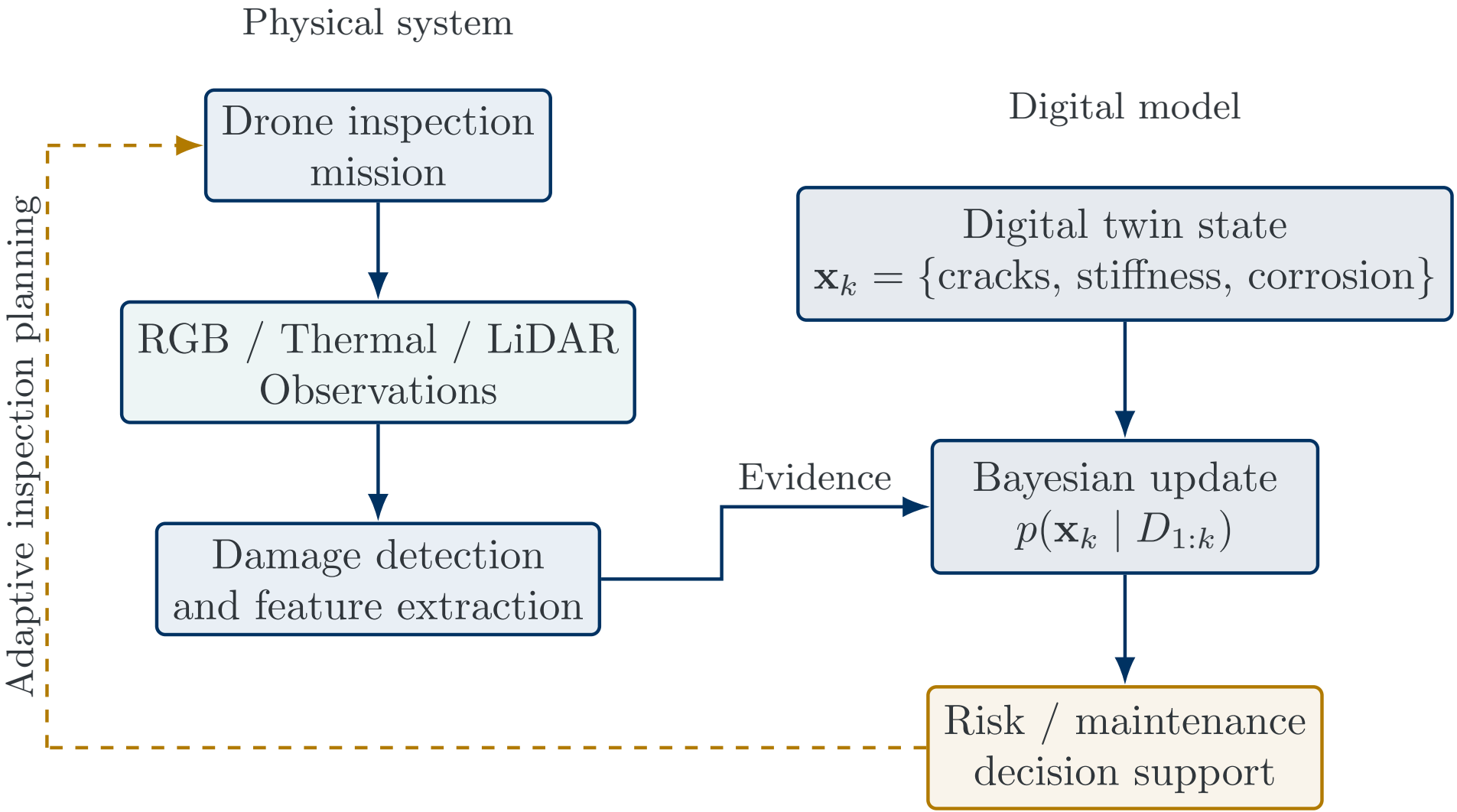

Physical system
Digital model
Drone inspection mission
RGB / Thermal / LiDAR Observations
Damage detection and feature extraction
Evidence
Digital twin state
$\mathbf{x}_k = \{\text{cracks, stiffness, corrosion}\}$
Bayesian update
$p(\mathbf{x}_k \mid D_{1:k})$
Risk / maintenance decision support
Adaptive inspection planning